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Physics 225 WFV
Problem Set 1

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Problem 1.1

Revisit our lecture on “relativity without light”. In one of the final steps, it was assumed that

$$\frac{A(v)^2 - 1}{v^2 A(v)^2} = \kappa > 0 \tag{1.1}$$

Explore the cases $\kappa = 0$ and $\kappa < 0$. Are these viable options? Why or why not?

Solution

Problem 1.2

To get the conditions above, we required closure— i.e., that the composition of two frame transformations $S \rightarrow S'$ and $S' \rightarrow S''$ is also a frame transformation, $S \rightarrow S''$. You will notice that we only used equality of the diagonal components of the composite transformation to get the equation in the problem above. What about the off diagonal components? Explore.

Solution

Problem 1.3

Given a basis for a vector space V , $\{\vec{e}_i\}, i = 1, \dots, n$, we considered the set of n dual vectors $\{\underline{e}^i\}$ according to the rule

$$\underline{e}^i[\vec{e}_j] = \delta^i_j \quad (1.2)$$

- (1) Show that this rule is sufficient to define the $\underline{e}^i$ as dual vectors. That is, we know how each $\underline{e}^i$ acts on all $v \in V$.
 - (2) Show that $\{\underline{e}^i\}$ is a basis for V^* .
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Solution

Problem 1.4

Show that if $\{\underline{e}^i\}$ is a basis for V^* and $\{\vec{e}_i\}$ is a basis for V , then $\{\vec{\mathbf{e}}_i \otimes \underline{e}^j \otimes \underline{e}^k\}$, $i, j, k = 1, \dots, n$, is a basis for the space of $(1, 2)$ tensors.

Solution

Problem 1.5

If g is a $(0, 2)$ tensor, then it defines an isomorphism between V and V^* , i.e., $g : V \rightarrow V^*$. Show that the inverse map $g^{-1} : V^* \rightarrow V$, such that $g^{-1}g = gg^{-1} = \text{Id}$, is a $(2, 0)$ tensor.

Solution

Problem 1.6

Starting from the definition of a tensor as a multilinear map, prove the standard transformation law for (k, l) -tensor components under a change of coordinates.

Solution

Problem 1.7

Show that the identity maps

$$\text{Id}_1 : V \rightarrow V, \text{Id}_1(v) = v, \forall v \in V \quad (2.1)$$

$$\text{Id}_2 : V^* \rightarrow V^*, \text{Id}_2(\omega) = \omega, \forall \omega \in V^* \quad (2.2)$$

Can be interpreted as the same $(1, 1)$ tensor. What are its components? What does this tensor do when it is fed a one-form and a vector? Does this remind you of any other tensorial operation?

Solution

References

- [1] Vercil Juan. Lecture Notes on General Relativity: Physics 225, August 2026. *Based on lectures delivered by Dr. Ian Vega, 1st Semester, AY 2026–2027.* URL: https://github.com/silvernuke911/Class_Notes/blob/main/Physics%20225/p225.pdf.

Appendices

Code

No code was used for this problem set